

University Of Michigan
AM Signal Recovery with Spectral Filtering
ECE550 - Project 1 Report – Winter 2026

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Date: 02/25/2026

Objective

This project aims to analyze the transmission and recovery of an amplitude modulated signal using Double Sideband Suppressed Carrier (DSB-SC) modulation. A baseband audio signal is modulated using a carrier frequency and then corrupted by additive noise to simulate a realistic communication channel. Frequency-domain analysis using the Fast Fourier Transform (FFT) and Power Spectral Density (PSD) estimation is applied to observe the spectral characteristics of the signal. A frequency-domain bandpass filter is designed to suppress noise and isolate the desired sidebands. Finally, coherent demodulation and low-pass filtering are used to reconstruct the original message signal, and the recovery performance is evaluated using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE).

Introduction

Amplitude Modulation (AM) remains one of the foundational techniques in communication systems, forming the basis for understanding more advanced modulation schemes used in modern wireless applications. Among its variants, Double Sideband Suppressed Carrier (DSB-SC) modulation provides improved power efficiency by transmitting only the information-bearing sidebands. However, like all analog communication methods, DSB-SC signals are highly sensitive to channel noise and interference, which can significantly degrade signal quality and distort the recovered message.

In practical communication systems, transmitted signals are often corrupted by additive noise, such as white Gaussian noise (AWGN), as well as undesired spectral components originating from interference sources. These disturbances spread across the frequency spectrum and overlap with the desired signal, making recovery challenging. Effective signal reconstruction therefore requires spectral analysis and filtering techniques capable of isolating the useful components from noise.

The Fourier Transform plays a critical role in this process by providing insight into the frequency-domain structure of modulated signals. Through Fast Fourier Transform (FFT) analysis, the spectral location of the carrier and sidebands can be identified, allowing the design of frequency-domain filters that selectively preserve the signal bandwidth while suppressing unwanted components. Power Spectral Density (PSD) estimation further aids in visualizing energy distribution and assessing noise impact.

In this project, a baseband audio signal is modulated using DSB-SC, corrupted by additive noise, and then recovered using spectral filtering and coherent demodulation. The performance of the recovery process is evaluated using Signal-to-Noise Ratio (SNR), Mean Squared Error (MSE), and PSD comparison. This study demonstrates the effectiveness of frequency-domain filtering techniques in restoring corrupted AM signals and reinforces fundamental concepts in modulation, spectral analysis, and signal recovery.

System Specifications

The communication system implemented in this project follows a Double Sideband Suppressed Carrier (DSB-SC) modulation scheme. The main parameters used during the simulation and analysis are summarized below.

The audio signal was processed using the sampling frequency provided by the audio file during loading in MATLAB, to simplify computation and maintain manageable data size, only the first few seconds of the audio signal were used.

The baseband message $m(t)$ consists of a normalized mono audio signal obtained by converting the original stereo audio to a single channel and removing its DC component. The carrier frequency used for modulation was $f_c = 10$ kHz.

The system uses Double Sideband Suppressed Carrier (DSB-SC) amplitude modulation, defined as $s(t) = m(t)\cos(2\pi f_c t)$

Two types of noise were introduced during the experiments Audio noise signal and Gaussian white noise combined with a narrowband interferer.

A frequency-domain bandpass filter was designed to isolate the sidebands around the carrier frequency while suppressing out-of-band noise components.

The signal recovery performance was evaluated using are the Signal-to-Noise Ratio (SNR), the Mean Squared Error (MSE) and the Power Spectral Density (PSD) comparison.

Methodology and Design Procedure

Generate / Load Baseband Signal

The first step consisted of loading a short audio signal to serve as the baseband message $m(t)$, (Load Baseband Signal). The audio was converted to a single channel and limited to three seconds to reduce computational complexity. To ensure proper spectral analysis, the DC component was removed, and the signal was normalized according to

$$m_{norm}(t) = \frac{m(t) - \mu_m}{\max(m(t))}$$

Where μ_m represents the mean value of the signal

The time-domain waveform was plotted to observe amplitude behavior, and the frequency-domain representation was obtained using the Fast Fourier Transform (FFT):

$$M(f) = \frac{m(t) - \mu_m}{\max(m(t))}$$

The magnitude spectrum $|M(f)|$ was used to estimate the effective baseband bandwidth B . This bandwidth plays a critical role in later filter design, since it determines how much of the sideband energy must be preserved during recovery.

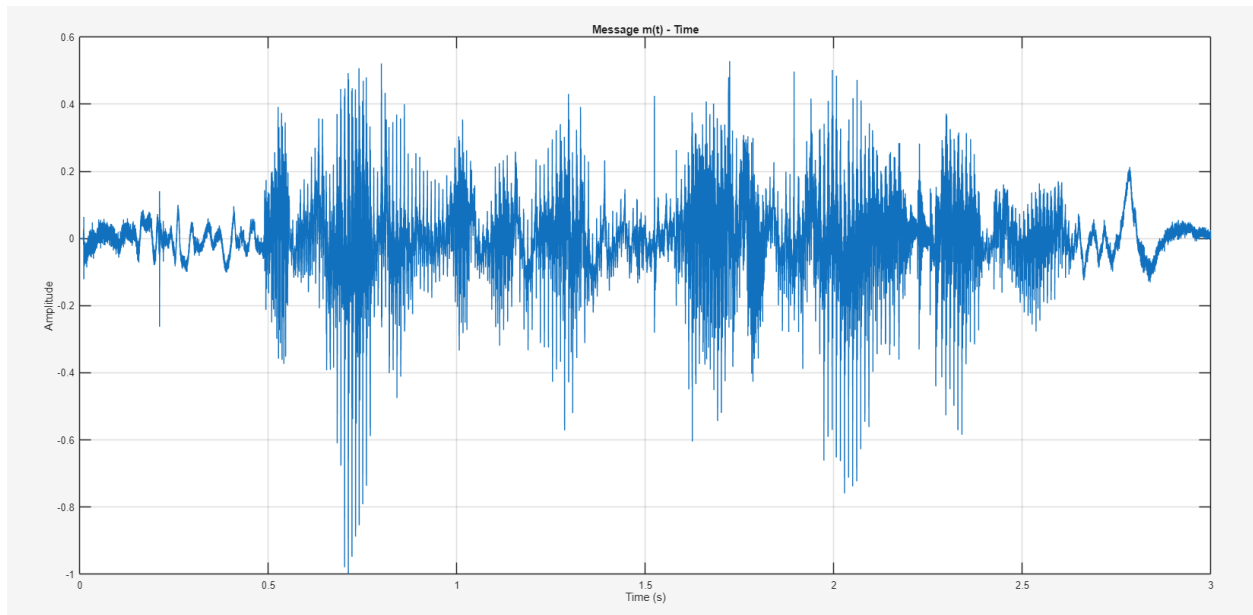


Figure 1: Message Audio Signal vs Time

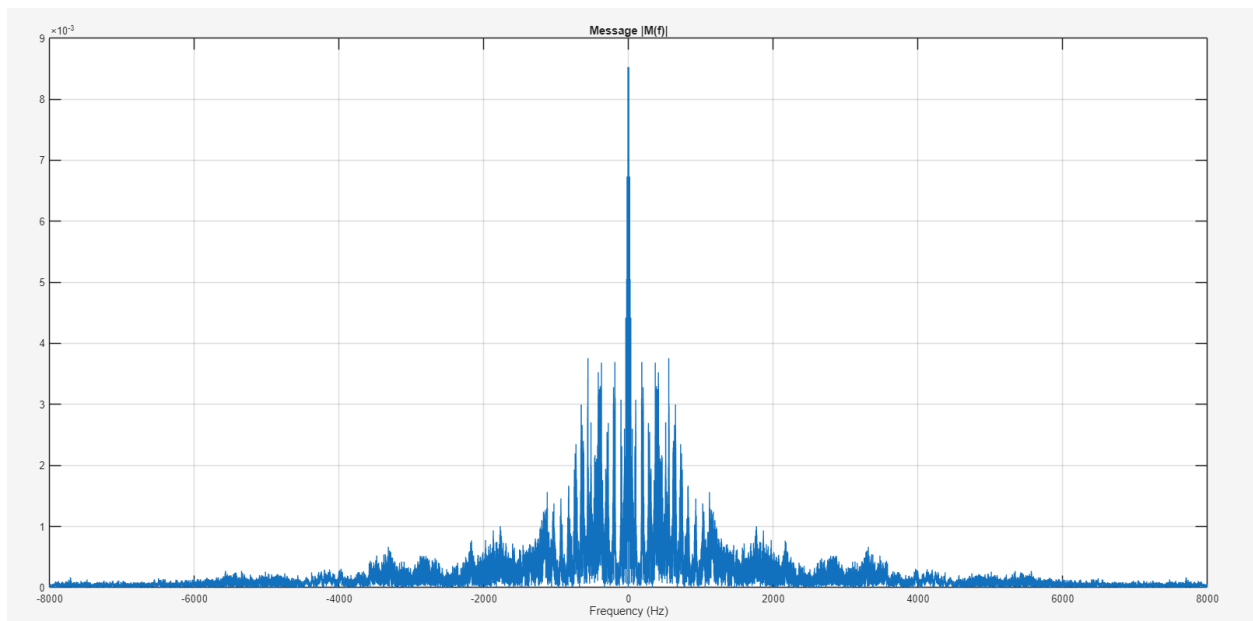


Figure 2: Magnitude Audio Signal vs Frequency

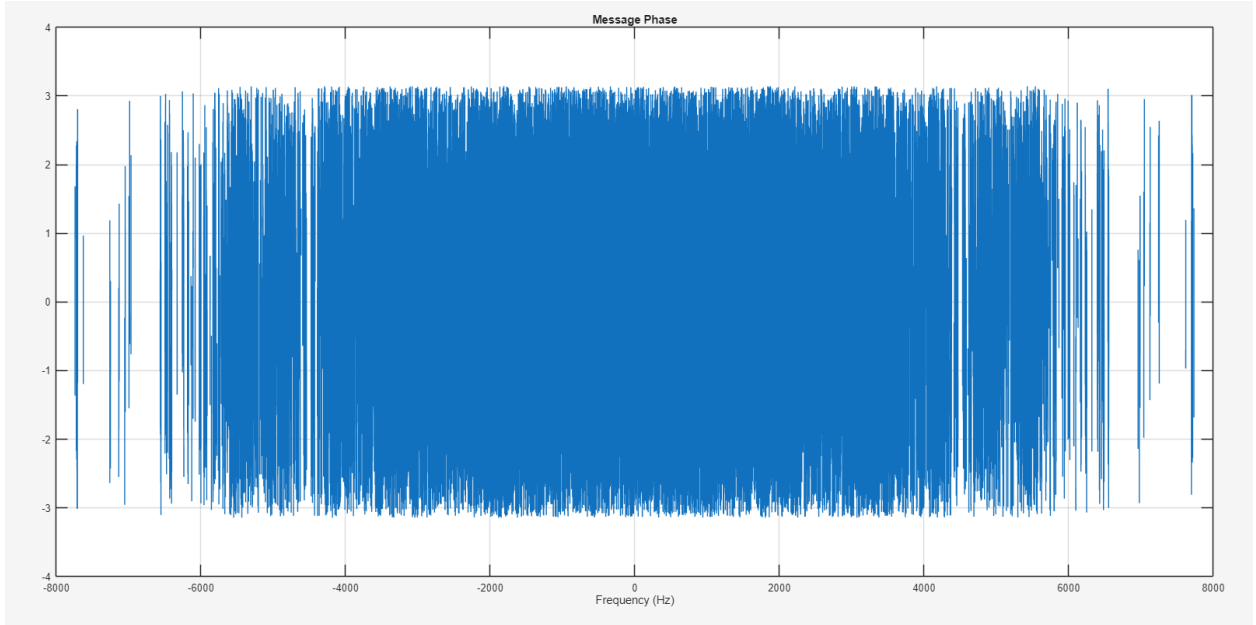


Figure 3: Message Phase vs Frequency

DSB-SC Modulation

In the second stage, the baseband message was modulated using Double Sideband Suppressed Carrier (DSB-SC) modulation. The modulated signal is defined as

$$s(t) = m(t) \cos(2\pi f_c t)$$

Where the carrier frequency was selected as $f_c = 10 \text{ kHz}$.

In the frequency domain, modulation shifts the spectrum of the baseband signal to frequencies centered around $\pm f_c$, yielding

$$s(t) = \frac{1}{2} [M(f - f_c) + M(f + f_c)]$$

As a result, the baseband spectrum is translated to two symmetric sidebands, and the total transmission bandwidth becomes $2B$. Spectral plots of the modulated signal confirm the presence of these sidebands around the carrier frequency.

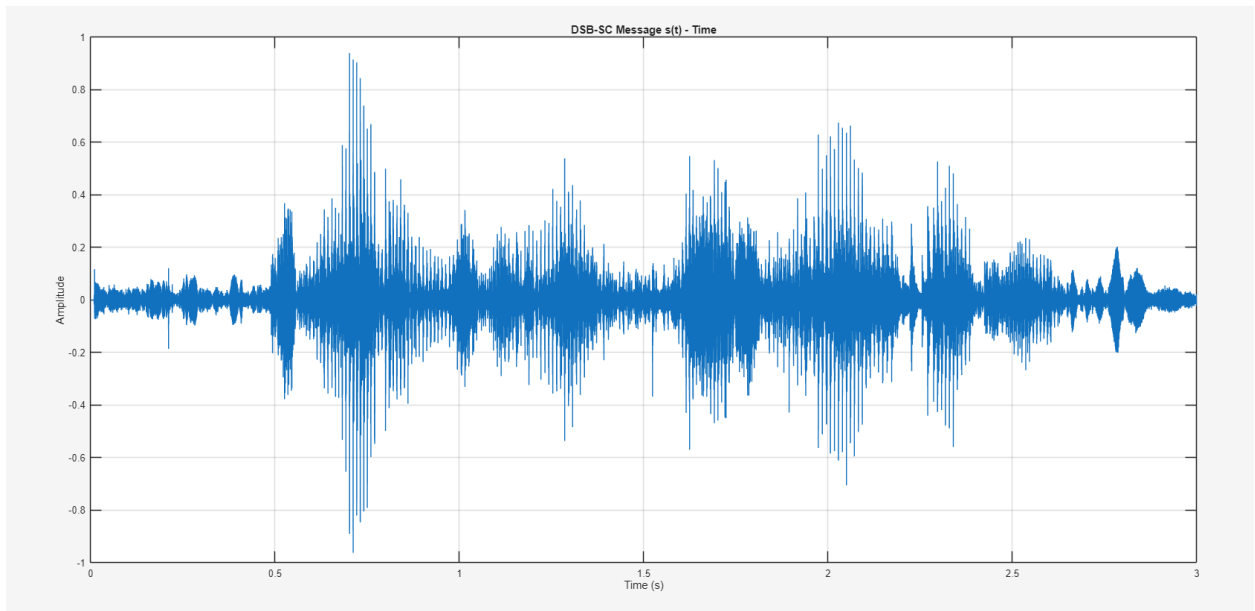


Figure 4: DSB-SC Message vs Time

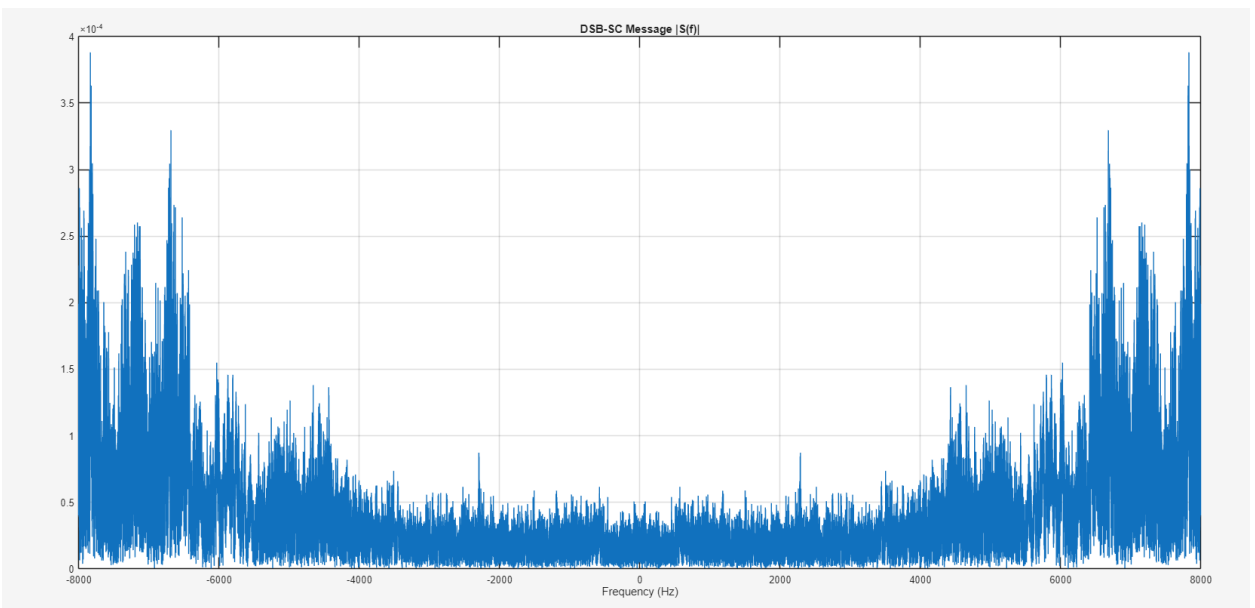


Figure 5: DSB-SC Message Magnitude vs Frequency

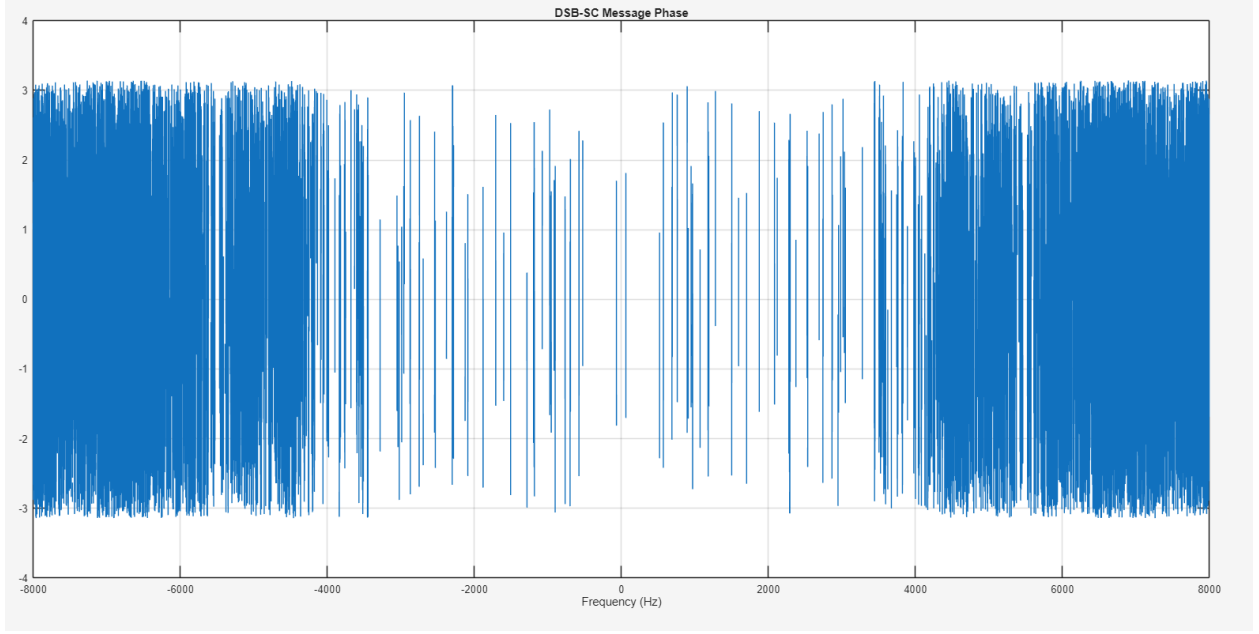


Figure 6: DSB-SC Message Phase vs Frequency

Addition of Noise

Following the workflow requirement to add noise and interference, the modulated signal was corrupted using audio noise. The noisy signal was constructed as

$$x(t) = s(t) + \alpha n(t),$$

where $n(t)$ is a zero-mean noise signal and α controls the noise amplitude. The noise signal was normalized prior to addition to ensure controlled signal-to-noise conditions.

In the frequency domain, the additive noise introduces a broadband spectral floor that overlaps the desired sidebands. FFT analysis of $x(t)$ clearly shows that although the carrier components remain visible, the sidebands are partially masked by noise. This degradation motivates the need for spectral filtering.

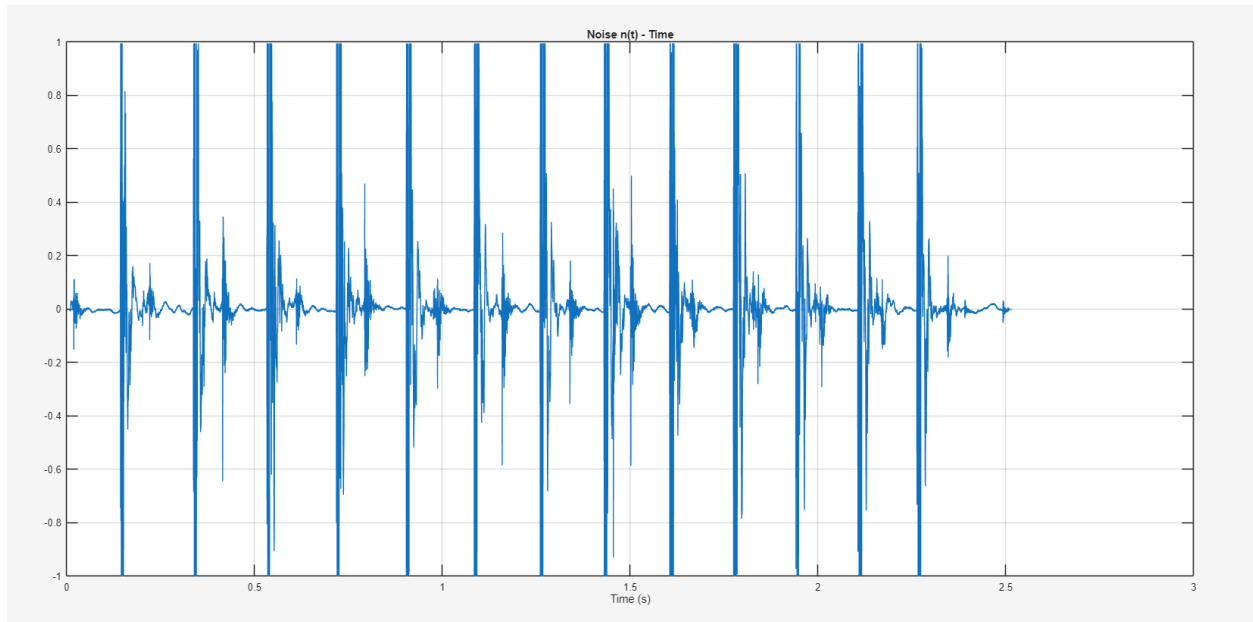


Figure 7: Noise vs Time

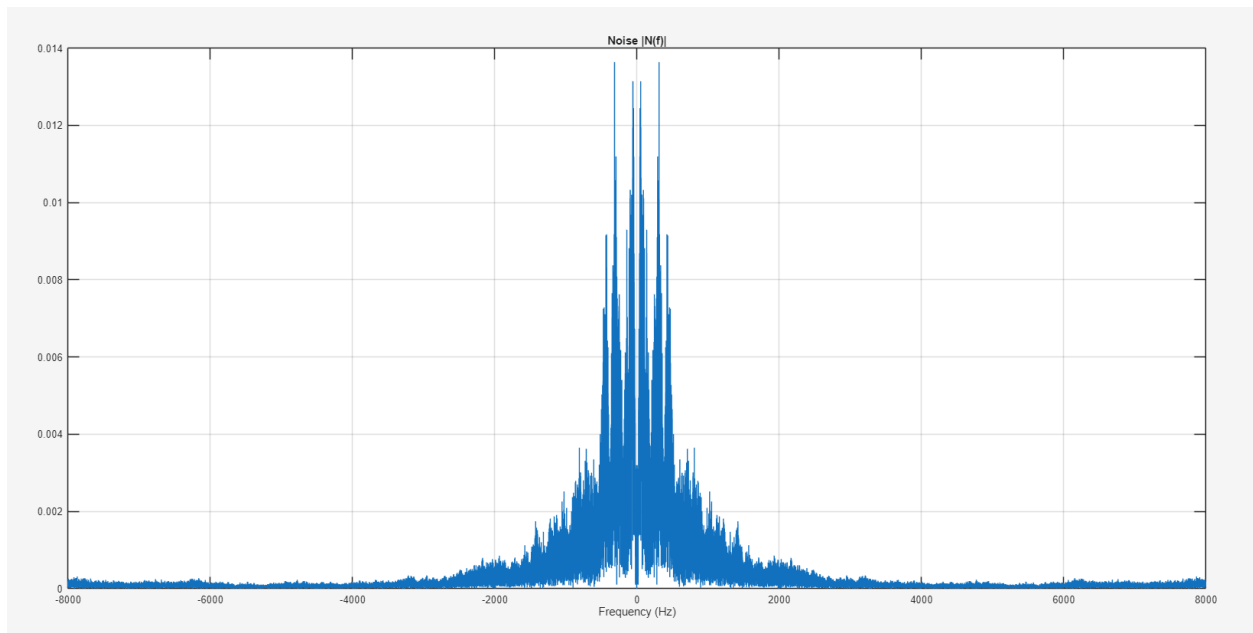


Figure 8: Noise Magnitude vs Frequency

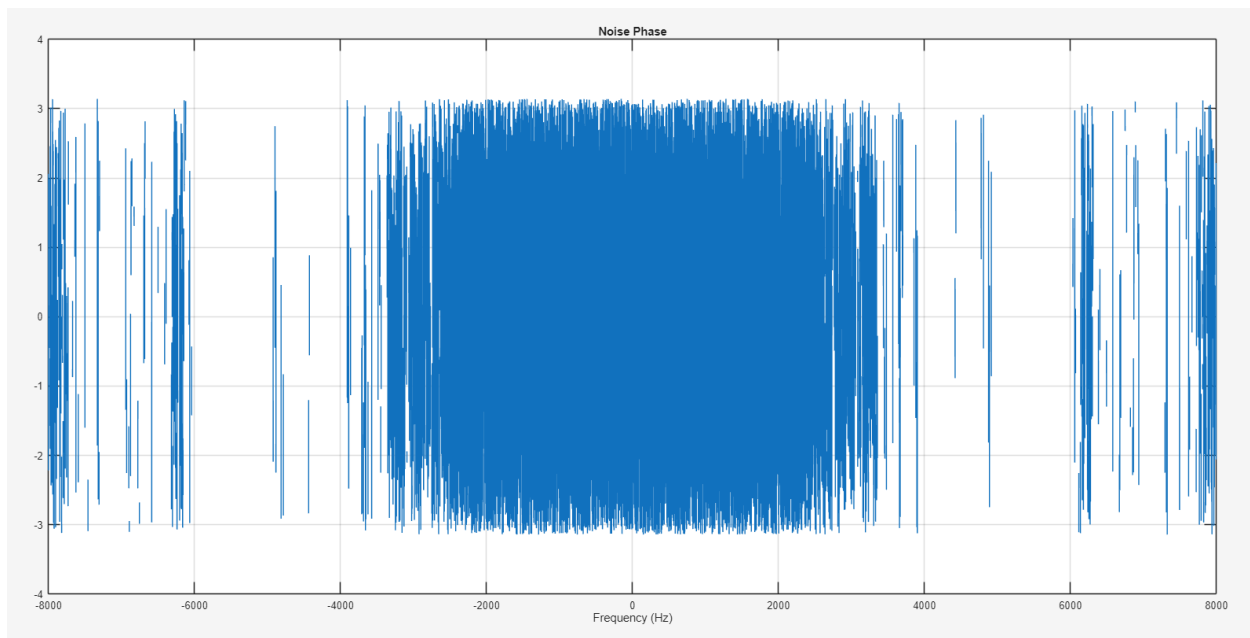


Figure 9: Noise Message Phase vs Frequency

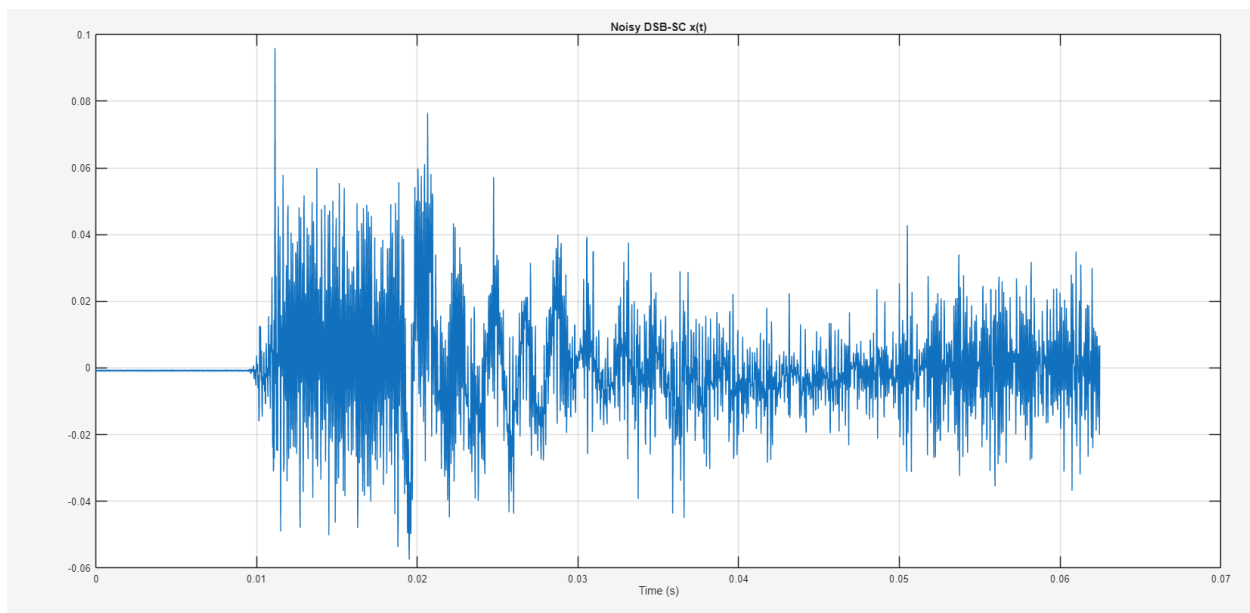


Figure 10: Noise and Message vs Time

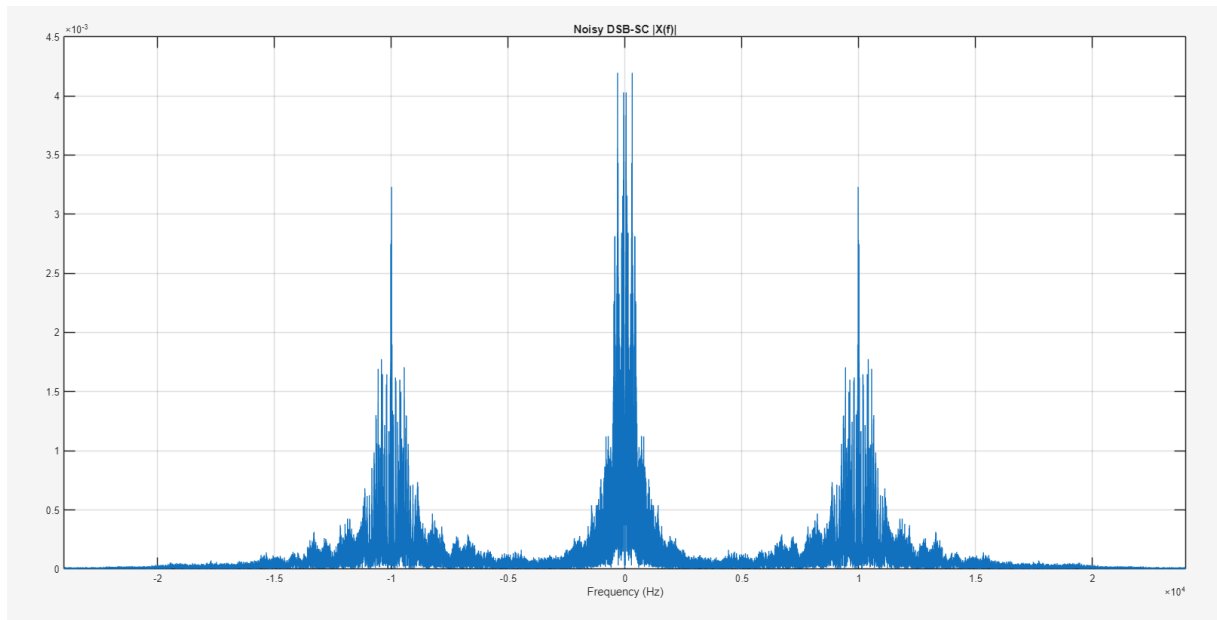


Figure 11: Noise and Message Magnitude vs Frequency

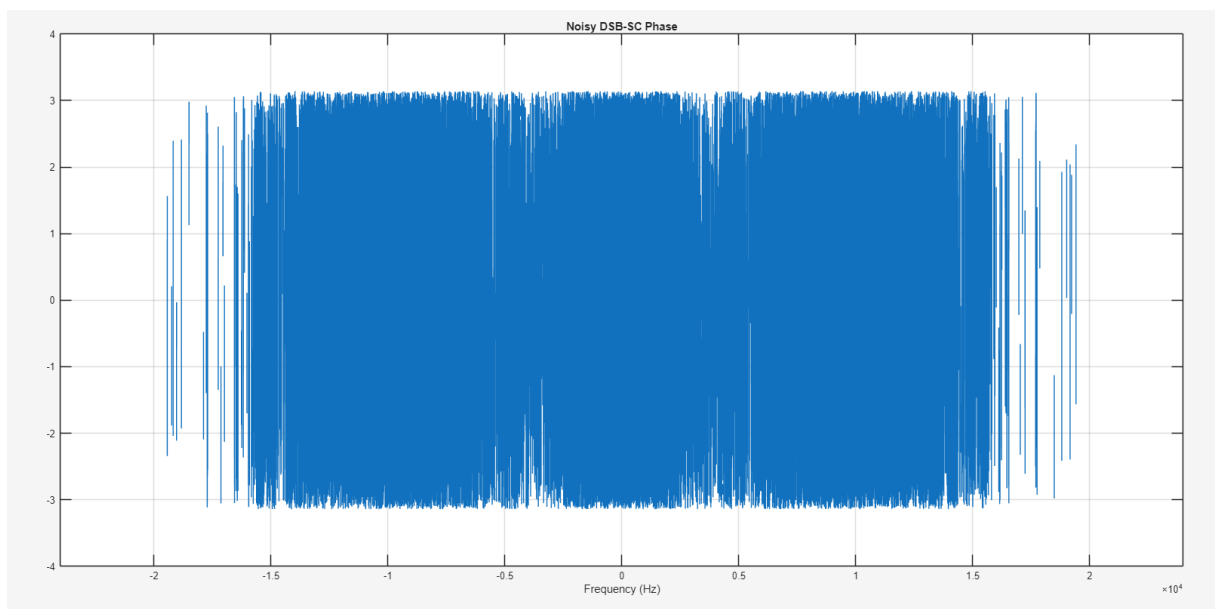


Figure 12: Noise and Message Phase vs Frequency

Frequency-Domain Analysis

To isolate the useful signal components, the corrupted signal was transformed into the frequency domain:

$$X(f) = \text{FFT}\{x(t)\}.$$

Inspection of $|X(f)|$ allows identification of the carrier locations at $\pm f_c$ and the corresponding sideband regions extending approximately $\pm B$ around each carrier. This spectral observation directly informs the design of the frequency-domain filter.

Power Spectral Density Estimation

To obtain a more statistically stable view of signal energy distribution, Welch's method was applied as required in the workflow. The PSD estimate is computed by averaging modified periodograms of overlapping windowed segments:

$$P_{xx}(f) = \frac{1}{K} \sum_{k=1}^K |\text{FFT}\{w_k(t)x(t)\}|^2.$$

Welch's method reduces variance compared to a single FFT estimate and provides a clearer visualization of noise floor levels and signal bandwidth. PSD plots were generated for the original message, the noise, and the corrupted modulated signal.

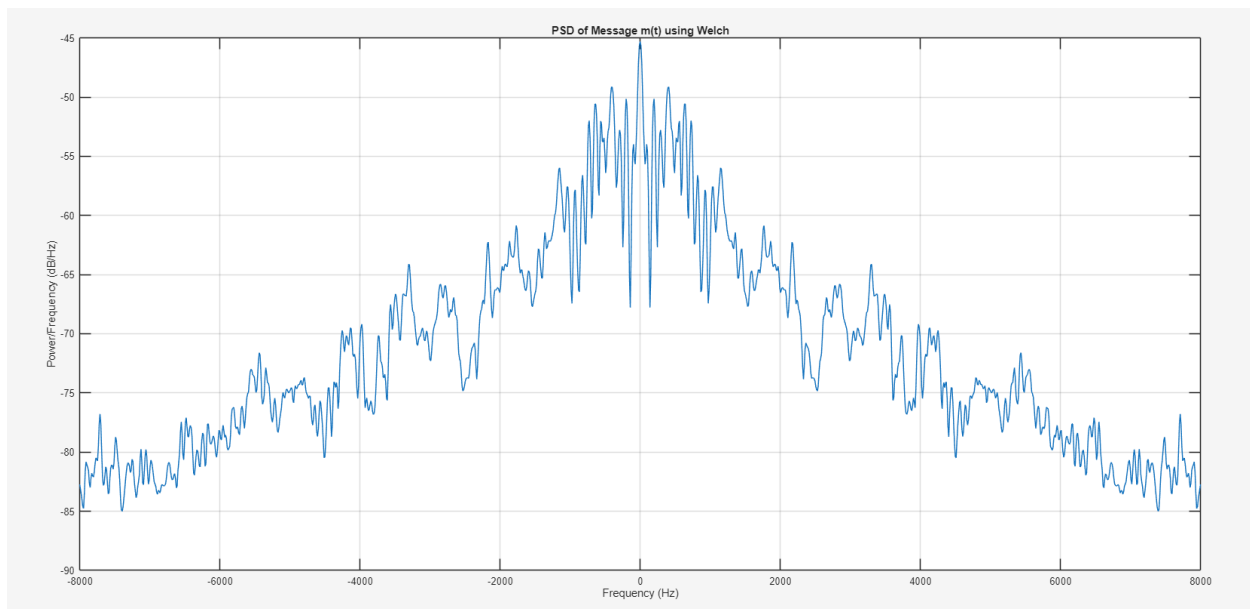


Figure 13: PSD of Message vs Frequency

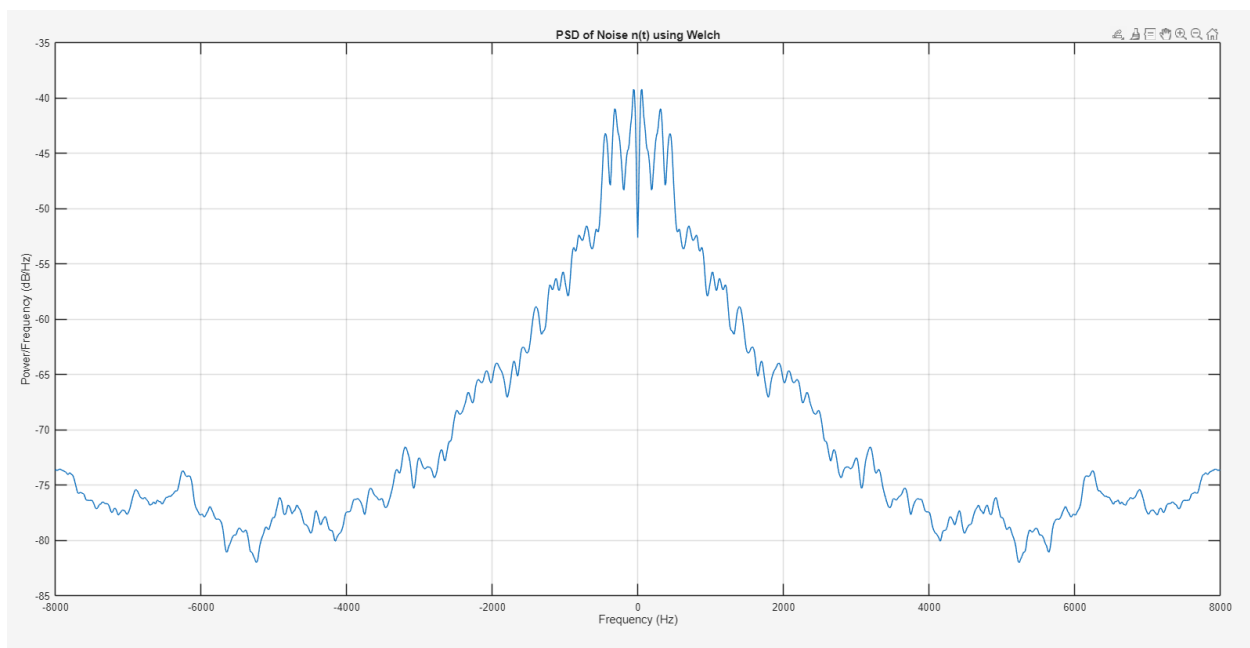


Figure 14: PSD of Noise vs Frequency

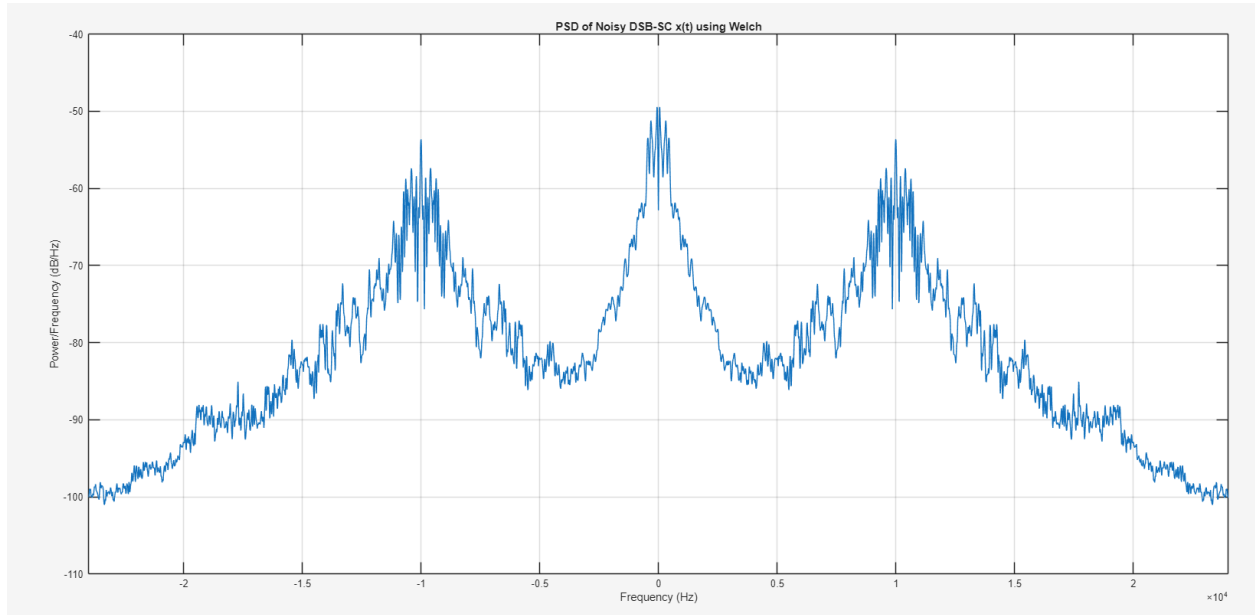


Figure 15: PSD of DSB-SC Noisy Message vs Frequency

Frequency-Domain Analysis

In accordance with Step 6 of the workflow, a bandpass filter was designed to isolate the carrier and its associated sidebands while suppressing out-of-band noise.

The baseband bandwidth B was estimated from the message spectrum. A safety margin was applied to prevent truncation of useful signal components.

A frequency-domain mask $H(f)$ was then constructed as

$$H(f) = \begin{cases} 1, & |f - f_c| \leq B \\ 1, & |f + f_c| \leq B \\ 0, & \text{otherwise.} \end{cases}$$

The filtered spectrum was obtained through pointwise multiplication:

$$Y(f) = X(f)H(f).$$

The corresponding time-domain filtered signal was computed using the inverse FFT:

$$y(t) = \text{IFFT}\{Y(f)\}.$$

This operation preserves only the spectral regions containing the desired DSB-SC components, effectively reducing broadband noise.

Coherent Demodulation and Baseband Recovery

After filtering, coherent demodulation was performed. The filtered signal was multiplied by a synchronized carrier:

$$z(t) = 2y(t)\cos(2\pi f_c t).$$

Using trigonometric identities, this operation produces

$$z(t) = m(t) + m(t)\cos(4\pi f_c t),$$

where the second term corresponds to a high-frequency component at $2f_c$. To recover the original message, a low-pass filter with cutoff frequency B was applied:

$$m_{rec}(t) = \text{LPF}\{z(t)\}.$$

The resulting signal represents the reconstructed baseband message.

Results and Performance Evaluation

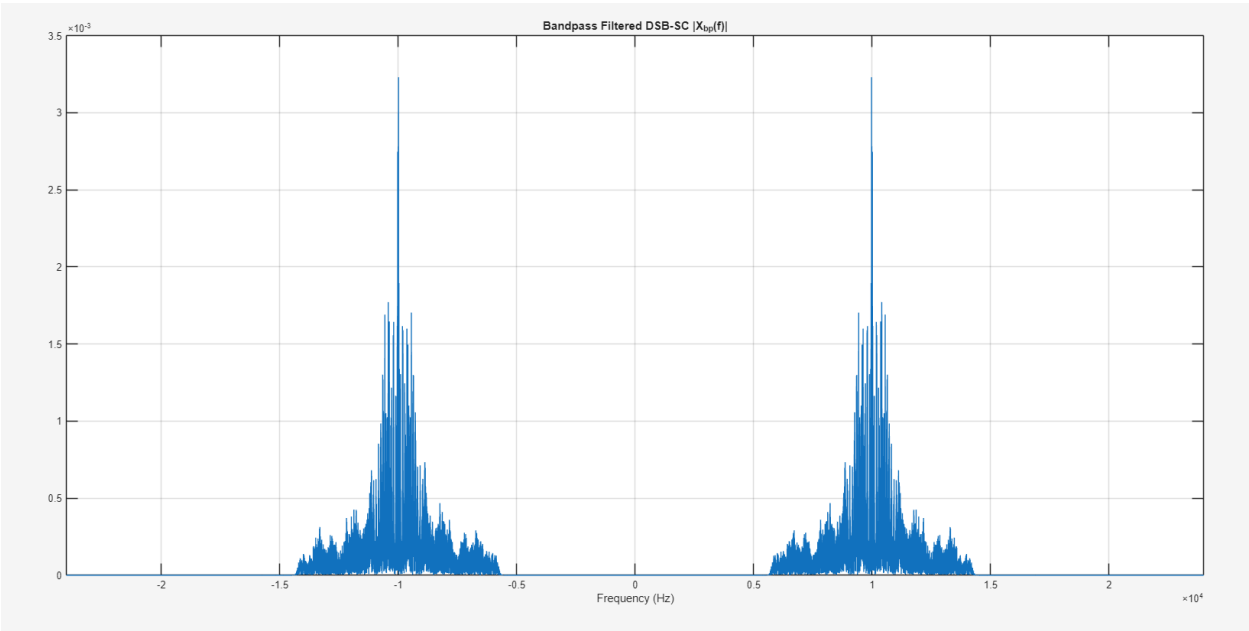


Figure 16: Bandpass Filtered DSB-SC Noisy Message vs Frequency

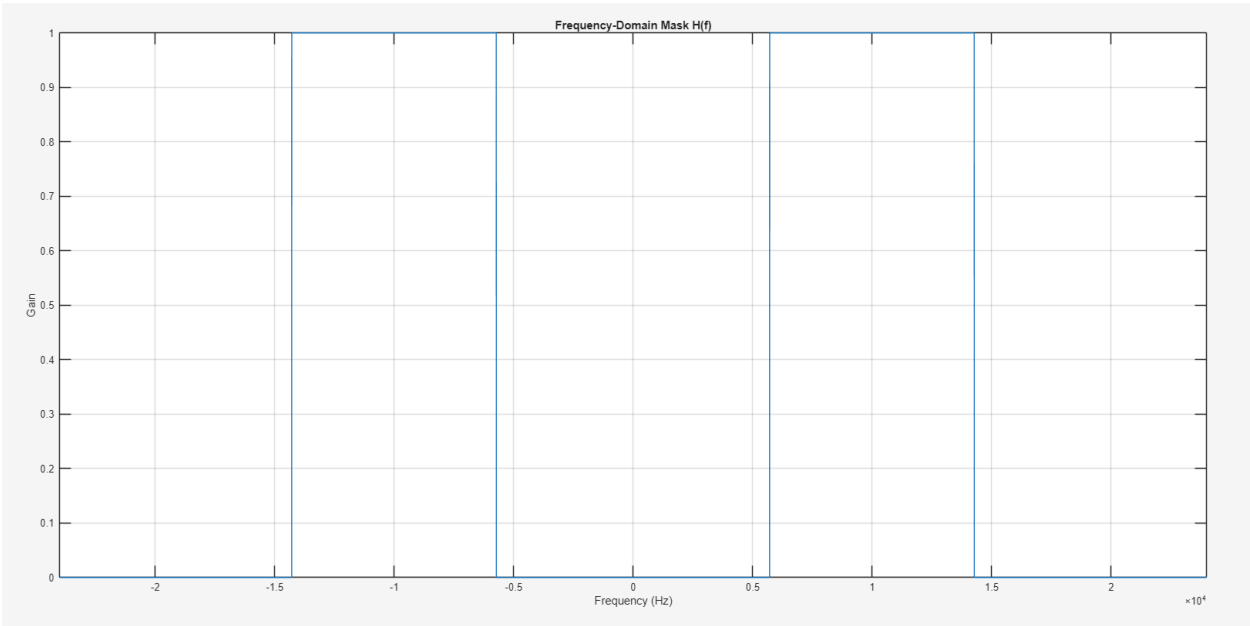


Figure 17: Frequency Domain Mask $H(f)$ vs Frequency

The final stage consisted of quantitatively evaluating the recovery performance. The Signal-to-Noise Ratio (SNR) was computed as

$$SNR = 10\log_{10} \left(\frac{\sum m^2(t)}{\sum (m(t) - m_{rec}(t))^2} \right),$$

The SNR according to the equation above equals 16.23 dB which is considered a good recovery.

and the Mean Squared Error (MSE) was calculated as

$$MSE = \frac{1}{N} \sum (m(t) - m_{rec}(t))^2.$$

The MSE according to the equation above equals 0.000249 which is considered a good recovery.

Additionally, the PSD of the recovered signal was compared with that of the original message to visually confirm spectral restoration.

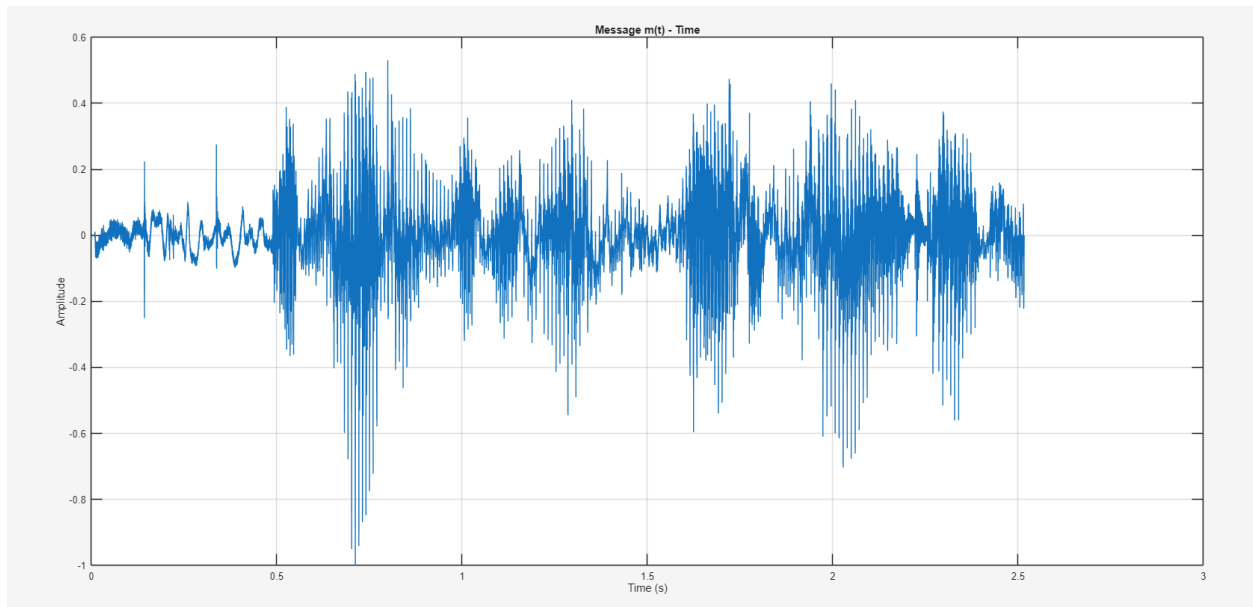


Figure 18: Recovered Message vs Time

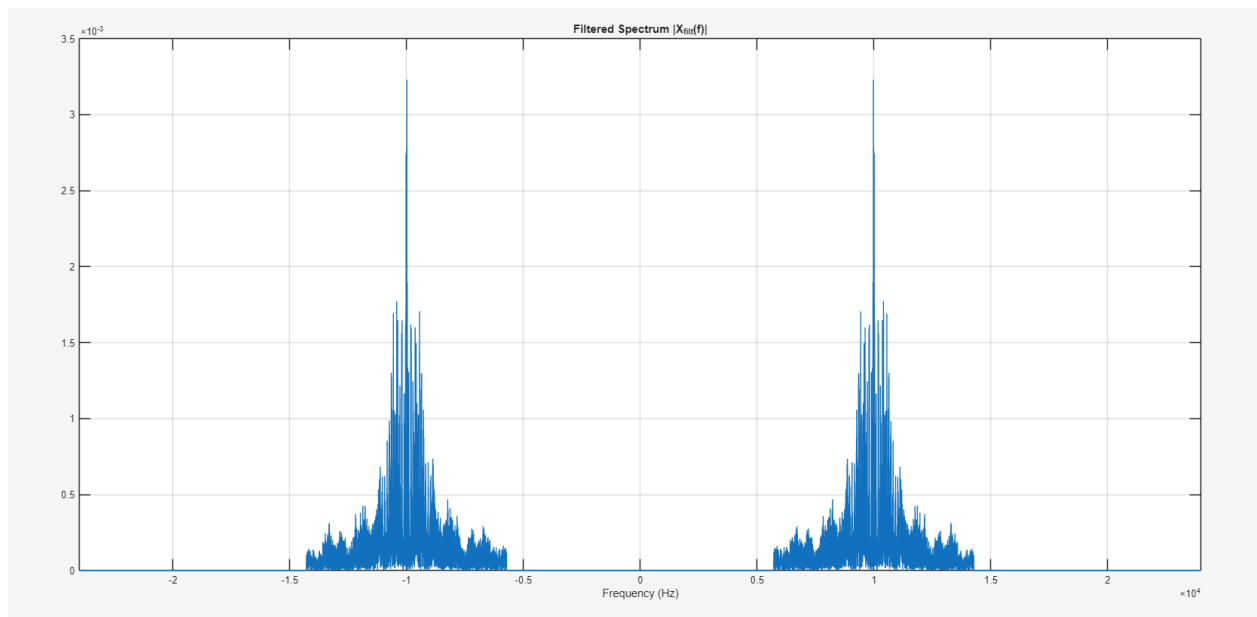


Figure 19: Filtered Noisy Message vs Frequency

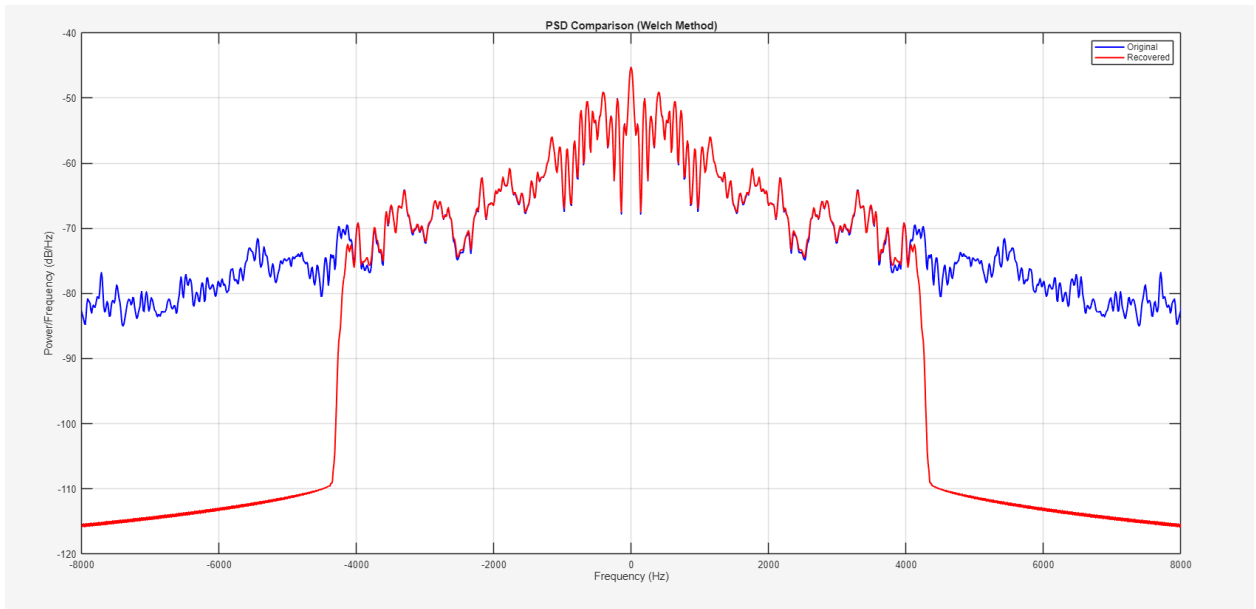


Figure 20: PSD Comparison vs Frequency

Additional Steps – Replace the audio noise with Gaussian noise and a narrowband interferer

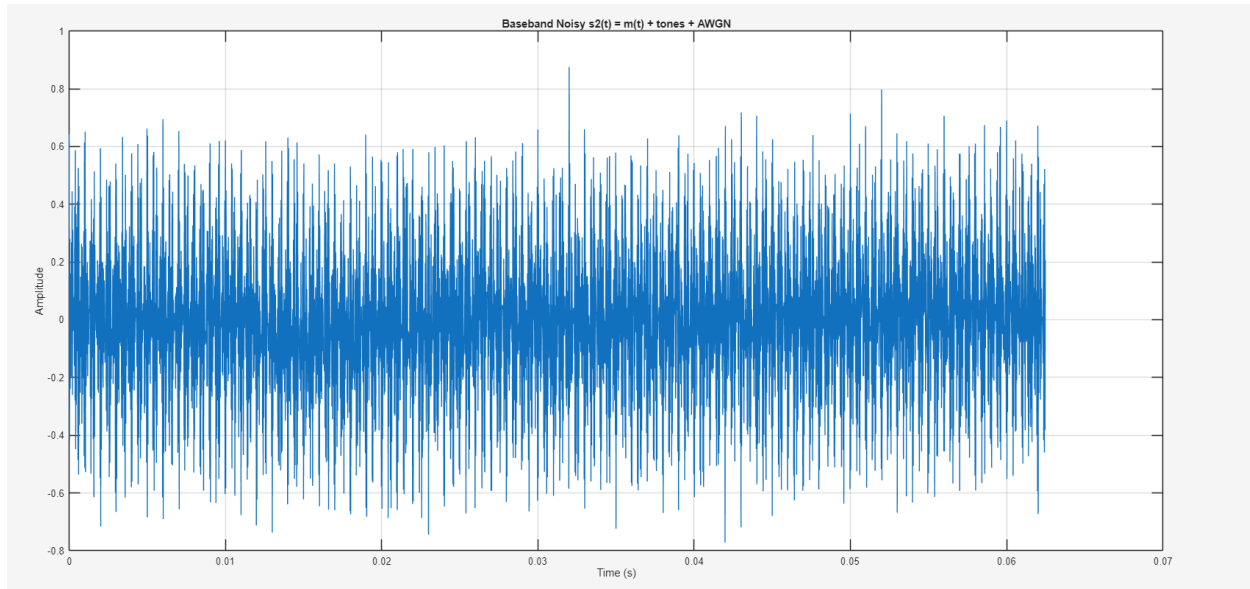


Figure 21: New Baseband Noise Message vs Time

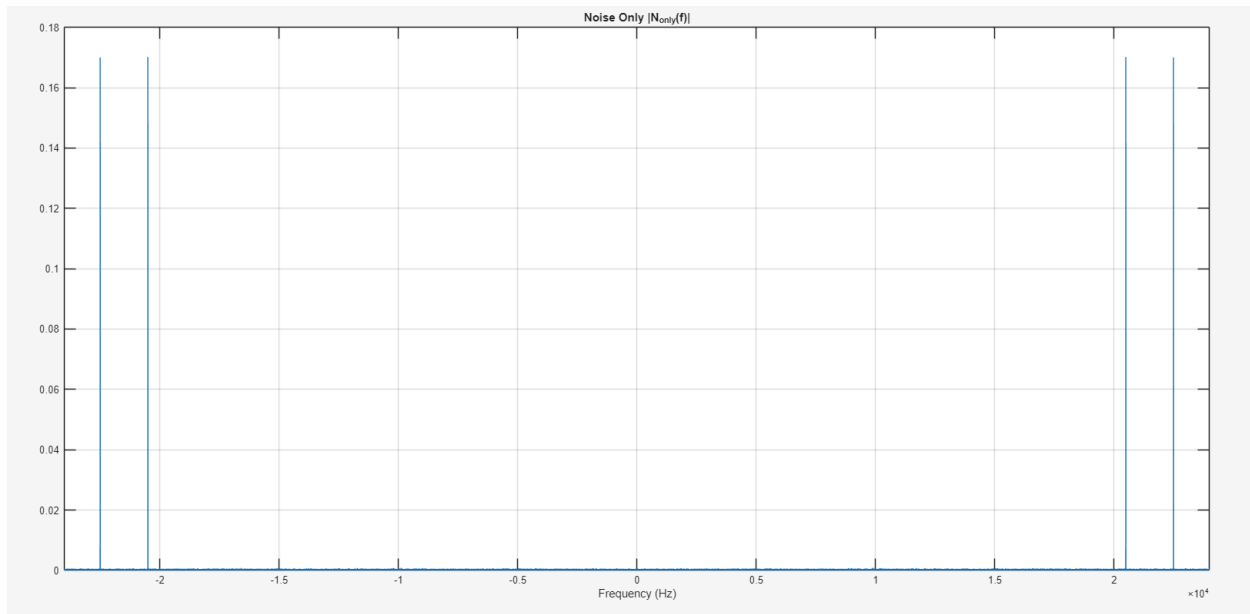


Figure 22: New Noise Message vs Frequency

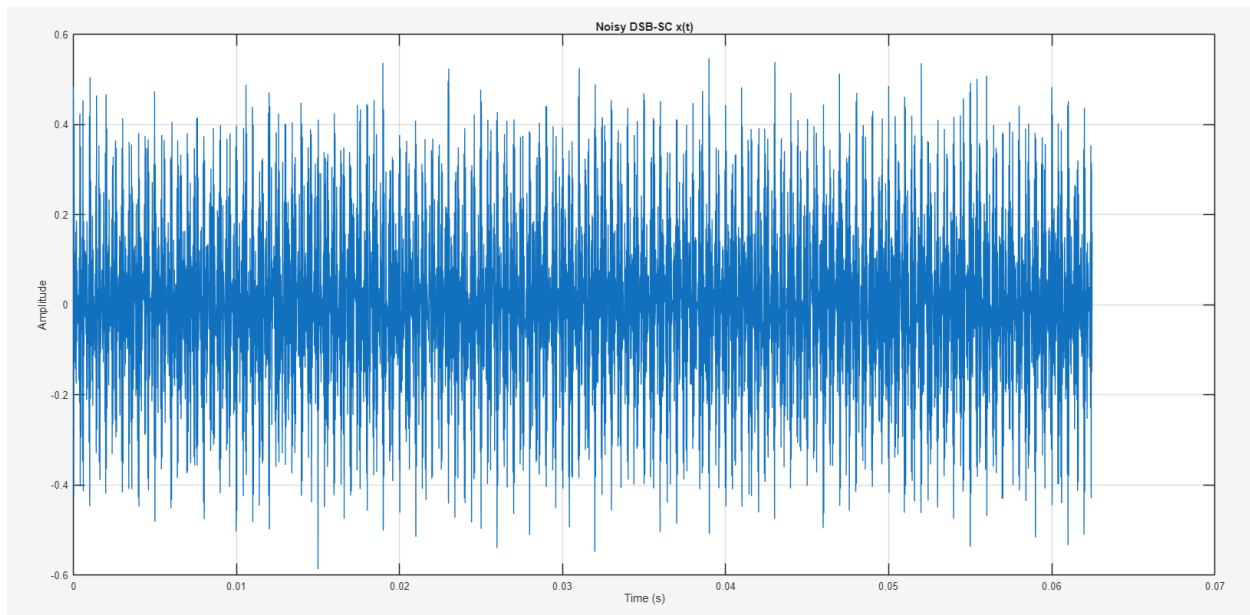


Figure 23: New Noisy DSB-SC Message

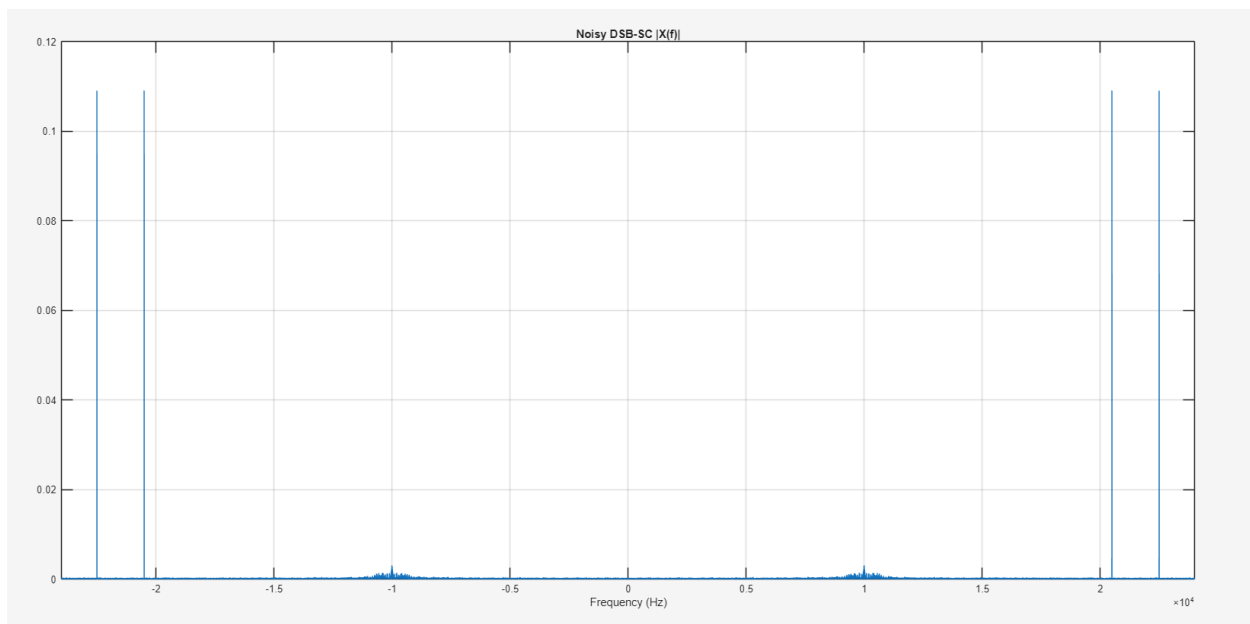


Figure 24: New Noisy DSB-SC Message phase

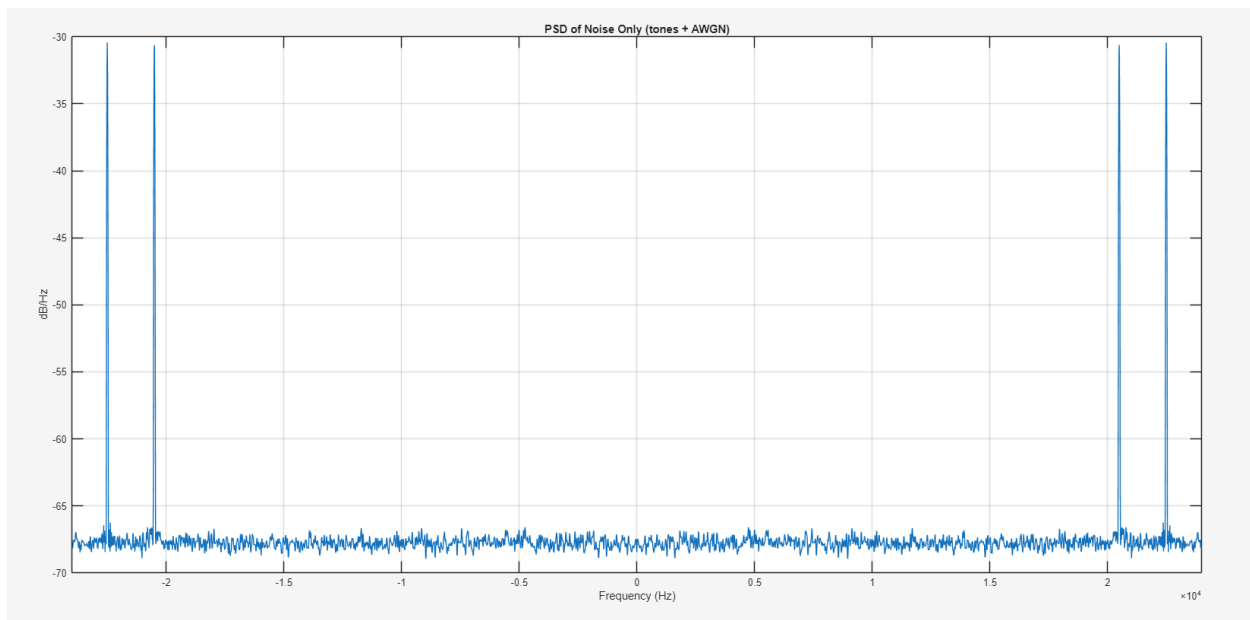


Figure 25: PSD of Noise only

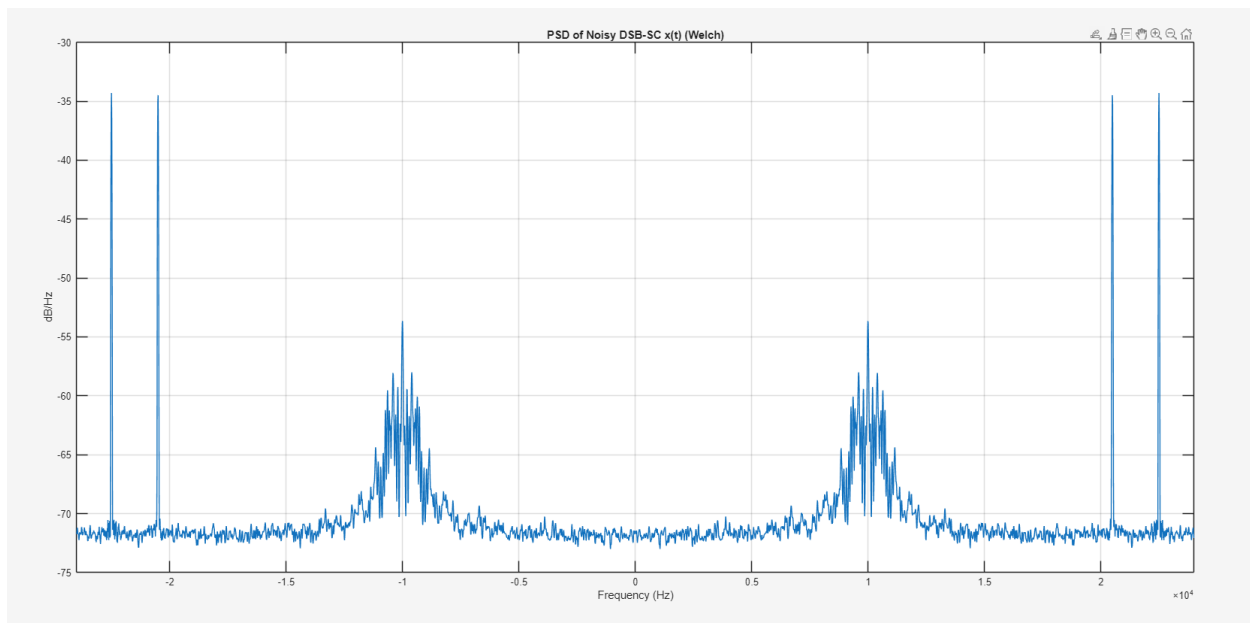


Figure 26: PSD of the New Noisy DSB-SC Message

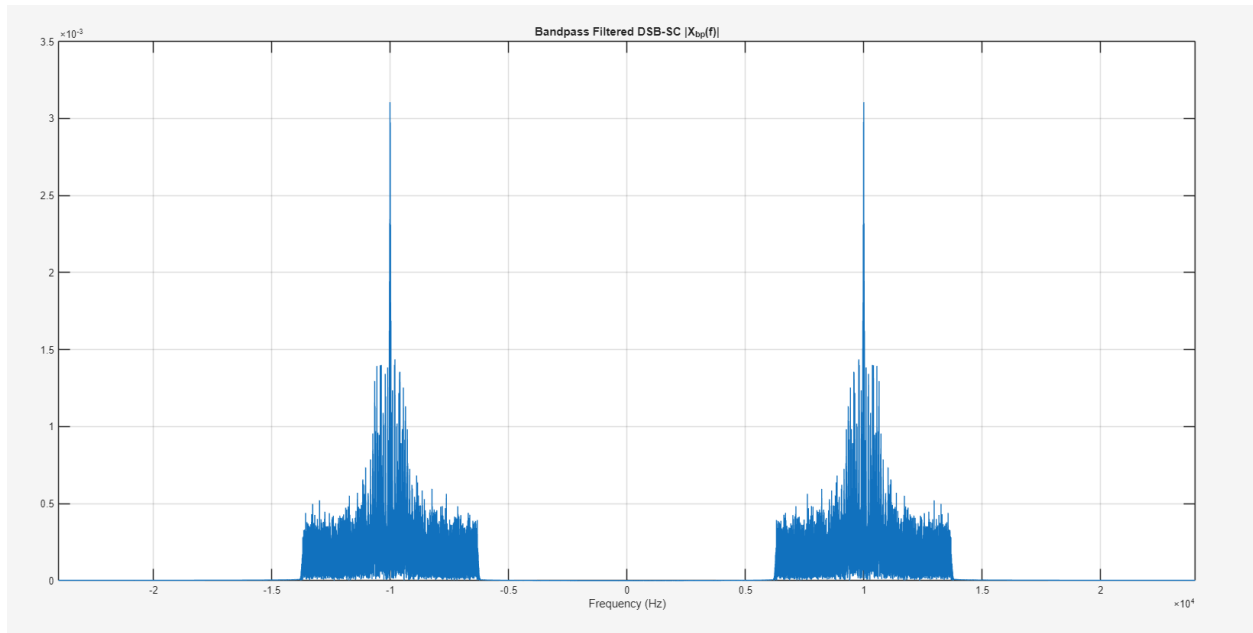


Figure 27: Bandpass filtered DSB-SC

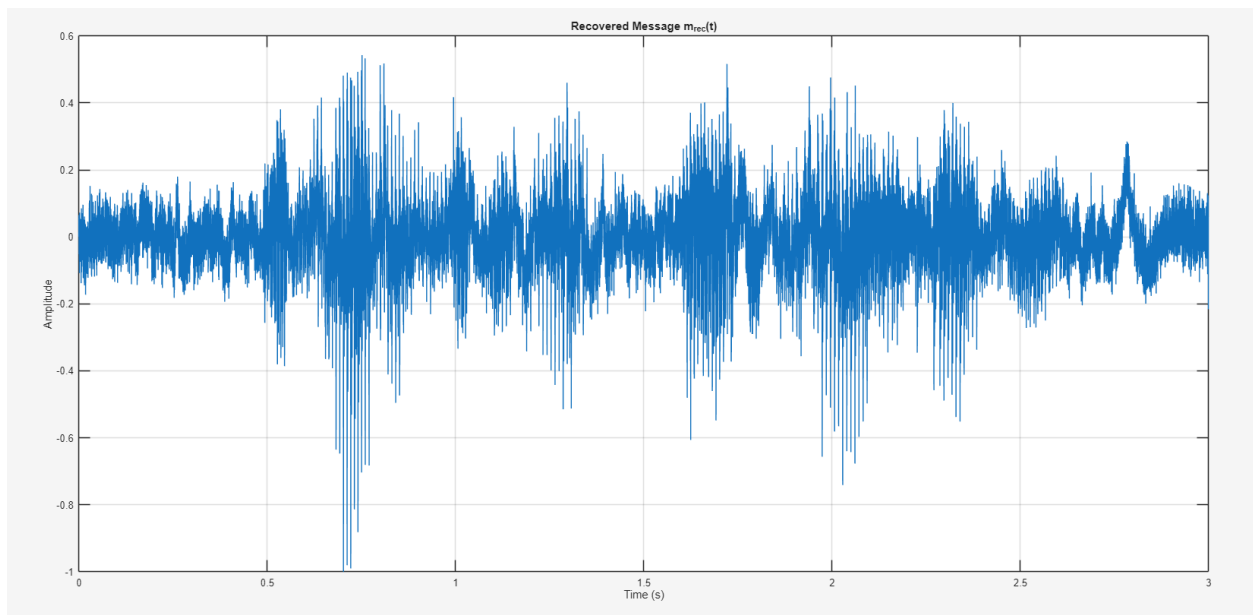


Figure 28: Recovered Message

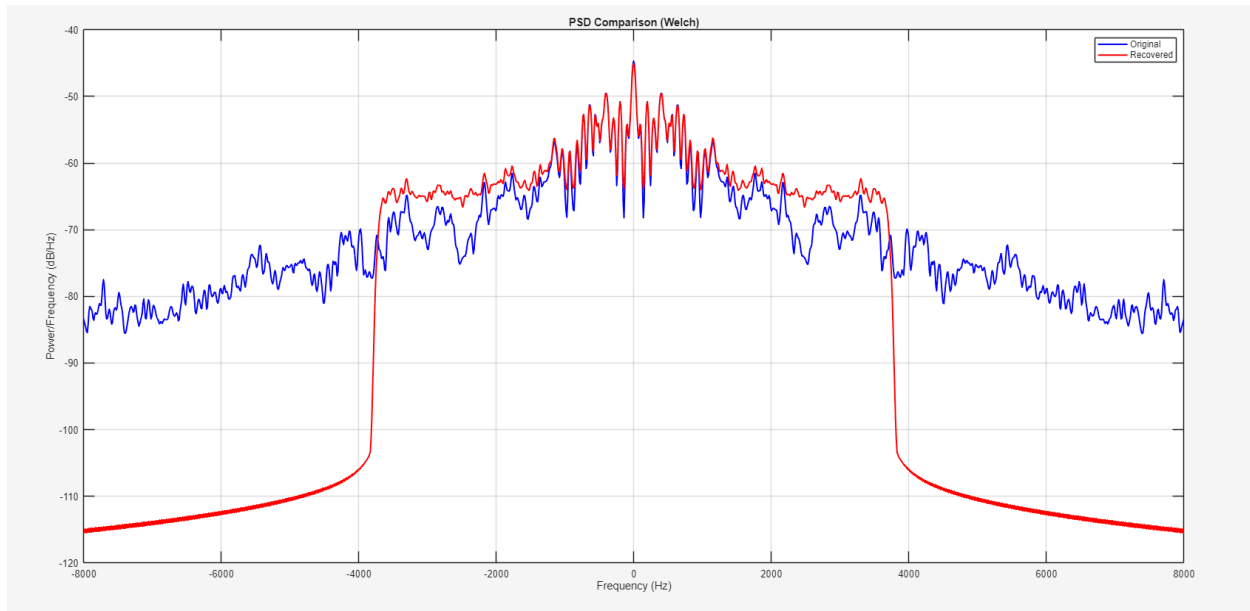


Figure 29: PSD comparison

The SNR according to the equation above equals 6.79 dB and the MSE equals 0.001967, that is considered as a poor recovery, that's because of the gaussian white noise that was added to the signal, and it is hard to remove and filtered.

Discussion

The results of this project demonstrate the effectiveness of frequency-domain spectral filtering in recovering a corrupted DSB-SC amplitude modulated signal. After the baseband audio signal was modulated with a 10 kHz carrier, the addition of noise significantly degraded the signal quality and obscured portions of the sideband information. In the frequency domain, this corruption appeared as an increased spectral floor and additional components overlapping the useful signal bandwidth.

Using FFT analysis, the location of the carrier and its corresponding sidebands was clearly identified. This information allowed the design of a bandpass filter that selectively preserved the spectral regions containing the modulated message while suppressing out-of-band noise components. The filtering process successfully reduced the noise energy and allowed the useful signal to be isolated before demodulation.

The comparison between FFT analysis and Power Spectral Density (PSD) estimation also highlights their different roles in signal analysis. The FFT provides a direct spectral representation of the signal and clearly shows the positions of the carrier and sidebands. However, FFT estimates can be sensitive to noise and spectral leakage. In contrast, PSD estimation using Welch's method provides a smoother representation of the signal's power distribution by averaging multiple spectral segments. This makes PSD particularly useful for observing the noise floor and identifying the effective signal bandwidth in noisy environments.

The recovery performance was evaluated quantitatively using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE). In the first experiment using audio noise, the recovered signal achieved an SNR of approximately 16.23 dB, indicating successful reconstruction of the original message. However, when Gaussian noise and a narrowband interferer were introduced, the SNR dropped to approximately 6.79 dB, demonstrating that white Gaussian noise is more difficult to suppress using simple bandpass filtering alone.

Overall, these results emphasize the importance of accurate bandwidth estimation and appropriate filter design in communication systems. While spectral filtering can significantly

improve signal quality, more advanced techniques such as adaptive filtering or optimal detection methods may be required when stronger noise or interference is present.

Conclusion

This project demonstrated the recovery of a corrupted DSB-SC amplitude modulated signal using frequency-domain spectral filtering techniques. Beginning with a baseband audio signal, DSB-SC modulation was applied to shift the spectrum to a carrier frequency of 10 kHz. The modulated signal was then corrupted by additive noise to simulate realistic channel conditions.

Frequency-domain analysis using the Fast Fourier Transform enabled clear identification of the carrier and sideband components. By designing a bandpass filter centered around the carrier frequency with bandwidth determined from the baseband spectrum, it was possible to suppress out-of-band noise while preserving the useful signal components. The filtered signal was then coherently demodulated and low pass filtered to reconstruct the original message.

Quantitative evaluation using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE) confirmed that spectral filtering significantly improved signal quality. Power Spectral Density (PSD) comparison further validated that the recovered signal closely matched the spectral characteristics of the original baseband message. Listening tests also supported the numerical results, demonstrating perceptible improvement after filtering.

Overall, the project reinforces fundamental concepts in Fourier analysis, modulation theory, bandwidth considerations, and spectral shaping. It illustrates how frequency-domain filtering serves as an effective tool for mitigating noise in communication systems and highlights the importance of proper bandwidth estimation and filter design in signal recovery applications.

Future Work

Several extensions and improvements can be explored to further enhance the performance and robustness of the signal recovery process. One possible improvement is the use of adaptive filtering techniques, such as Wiener filtering or LMS adaptive filters, which can dynamically adjust filter parameters based on the noise characteristics and improve noise suppression compared to a fixed bandpass filter.

Another potential extension is the implementation of digital FIR or IIR filters instead of a simple frequency-domain mask. Carefully designed filters can provide better control over passband ripple and stopband attenuation, leading to improved signal reconstruction quality.

Future experiments could also investigate the performance of the recovery system under different channel conditions, including higher levels of Gaussian noise, colored noise, or multiple interference sources. This would provide a more realistic evaluation of the system's robustness.

Additionally, the project could be extended to analyze other modulation schemes such as Single Sideband (SSB) or Quadrature Amplitude Modulation (QAM) to compare their spectral efficiency and noise resilience with DSB-SC modulation.

Finally, a real-time implementation using MATLAB real-time processing or software-defined radio (SDR) platforms could be explored to demonstrate practical communication system applications.

References

- Textbooks: Rodger E. Ziemer and William Tranter, Principles of Communications
- Systems, Modulation and Noise, Sixth Edition
- Simon Haykin and Michael Moher, Introduction to Analog & Digital Communications,
- Second Edition, Wiley
- Computer
- Matlab